

RODINO, Russian Federation

WMO: 360200

Lat: 52.486N

Lon: 80.183E

Elev: 159

StdP: 99.43

Time Zone: 7.00 (E07)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-36.4	-33.6	-39.2	0.1	-36.3	-36.1	0.1	-33.6	13.3	-4.5	12.1	-6.3	0.9	200	0.621

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	31.3	18.4	29.6	17.9	27.8	17.5	20.3	27.6	19.5	26.5	18.7	25.4	3.9	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.9	13.1	22.7	17.0	12.4	22.1	16.2	11.7	21.8	59.3	27.7	56.5	26.5	53.9	25.5	24.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.9	10.1	8.9	-37.3	33.8	4.5	2.6	-40.6	35.6	-43.2	37.1	-45.7	38.6	-49.0	40.4
			-37.4	22.1	4.5	1.1	-40.6	22.9	-43.2	23.6	-45.7	24.2	-49.0	25.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.0	-17.6	-15.7	-6.0	6.5	12.5	19.5	20.5	17.8	11.5	3.8	-5.4	-12.8
	DBStd	14.73	8.94	7.84	6.70	5.89	5.17	4.24	3.25	3.54	4.78	5.04	7.87	8.26
	HDD10.0	3649	857	719	495	133	35	1	0	0	37	201	463	708
	HDD18.3	5815	1115	953	753	354	192	38	17	52	212	450	713	966
	CDD10.0	1081	0	0	0	29	111	284	324	242	82	9	0	0
	CDD18.3	206	0	0	0	1	9	72	82	36	7	0	0	0
	CDH23.3	2072	0	0	0	13	124	668	727	416	123	0	0	0
	CDH26.7	622	0	0	0	1	23	215	228	112	42	0	0	0
Wind	WSAvg	3.7	3.7	3.7	4.0	4.1	4.2	3.3	2.9	3.0	3.2	3.5	4.2	4.4
Precipitation	PrecAvg	329	16	13	17	20	26	38	58	37	25	28	29	22
	PrecMax	426	27	28	40	49	79	72	108	97	54	77	48	39
	PrecMin	182	4	1	4	0	8	8	16	3	3	9	11	10
	PrecStd	54	7	7	9	13	16	19	28	19	14	16	10	8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.5	1.2	11.0	25.3	29.7	33.0	33.6	32.8	31.3	21.5	11.1	2.1
		MCWB	0.0	-0.1	6.5	13.7	15.9	18.8	20.2	18.9	17.3	10.7	7.9	1.0
	2%	DB	-1.0	-0.8	6.4	21.9	26.6	31.0	31.1	29.6	27.2	17.3	8.0	0.7
		MCWB	-1.8	-1.9	3.1	11.8	14.4	18.0	19.5	17.8	14.9	10.1	5.5	-0.1
	5%	DB	-3.7	-2.4	4.0	18.9	24.4	29.2	29.2	27.3	23.4	14.2	5.5	-0.6
		MCWB	-4.5	-3.4	2.0	10.3	13.6	17.2	19.0	16.9	13.6	8.6	3.3	-1.5
	10%	DB	-5.5	-4.4	2.3	15.9	22.0	27.3	27.4	25.5	20.3	11.3	3.6	-2.1
		MCWB	-6.2	-5.3	0.7	9.0	12.6	17.2	18.5	16.5	12.3	7.4	2.0	-2.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.2	0.1	6.6	14.6	17.2	20.5	21.6	20.5	19.2	12.3	8.3	1.2
		MCDB	1.6	1.0	10.7	23.7	25.1	27.6	29.6	29.5	29.7	17.3	10.6	1.8
	2%	WB	-1.8	-1.7	3.4	12.7	15.7	19.6	20.8	18.9	15.9	10.8	5.6	0.1
		MCDB	-0.9	-0.7	5.8	20.3	24.1	27.0	28.1	26.6	23.3	16.1	7.7	0.7
	5%	WB	-4.4	-3.4	2.1	11.0	14.4	18.8	20.0	18.1	14.4	9.1	3.4	-1.4
		MCDB	-3.7	-2.5	3.8	17.3	22.2	26.4	26.8	25.2	22.0	14.4	5.3	-0.8
	10%	WB	-6.3	-5.3	0.8	9.6	13.2	18.1	19.2	17.2	12.8	7.6	1.8	-2.9
		MCDB	-5.5	-4.4	2.3	15.4	21.0	25.5	25.6	24.0	19.5	10.9	3.4	-2.2
Mean Daily Temperature Range	5% DB	MDBR	7.0	8.2	8.1	10.5	11.8	10.8	10.3	11.6	10.9	7.5	5.8	7.2
		MCDBR	6.8	6.6	8.5	14.7	14.6	13.5	13.3	14.9	16.2	12.1	7.2	6.6
		MCWBR	6.7	6.4	6.2	7.6	6.5	4.5	4.7	6.1	7.1	6.7	5.2	6.7
	5% WB	MCDBR	6.8	6.7	7.9	12.9	12.4	10.7	10.8	12.6	13.7	10.7	6.7	6.4
		MCWBR	6.8	6.5	5.9	7.4	6.6	4.5	4.4	5.6	6.9	6.5	5.1	6.5
Clear-Sky Solar Irradiance	taub	0.300	0.336	0.389	0.410	0.414	0.414	0.438	0.417	0.375	0.354	0.339	0.308	
	taud	2.022	1.949	1.915	2.141	2.181	2.269	2.200	2.225	2.288	2.244	2.087	1.980	
	Ebn at Noon	643	731	770	819	836	840	809	804	796	723	589	541	
	Edh at Noon	80	117	150	136	137	127	134	124	103	87	76	71	
All-Sky Solar Radiation	RadAvg	1.21	2.32	3.79	4.73	5.94	6.35	5.76	5.02	3.66	1.96	0.98	0.82	
	RadStd	0.10	0.12	0.31	0.34	0.43	0.42	0.39	0.33	0.31	0.27	0.12	0.12	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page